THE BIOROBOTICS INSTITUTE

Scuola Superiore Sant'Anna

Master of Science in Bionics Engineering

Academic Year 2025-2026

Pontedera, March 5, 2026

Bioinspired and Soft Robotics

Soft Robotics Technologies

Embodiment
and Morphological Computation

Matteo Cianchetti - matteo.cianchetti@santannapisa.it

Introduction to embodiment

"the behavior of any system is not merely the outcome of an internal control structure (such as the central nervous system). A system's behavior is also affected by the ecological niche in which the system is physically embedded, by its morphology (the shape of its body and limbs, as well as the type and placement of sensors and effectors), and by the material properties of the elements composing the morphology."

(Pfeifer et al., 2007)

TRIVIAL

intelligence requires a body

DEEPER IMPLICATIONS

the impact of physical processes on informational processes

the implications of embodiment for cognition

Ground for the "embodied turn"

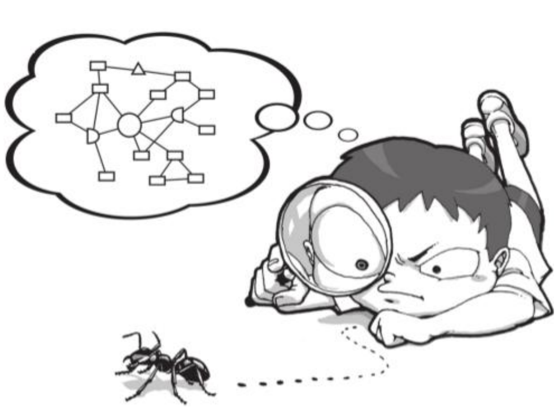
<u>Synthetic methodology</u>
the potential of the synthetic approach to studying intelligence: "understanding by building"

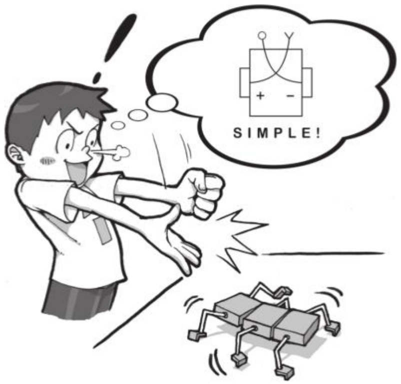
<u>Frame of reference problem</u>
too often, we automatically attribute intelligence to a certain behavior, without realizing that the mechanism that brings it about may be surprisingly simple

Synthetic methodology

ANALYTIC APPROACH

The analytic approach is universally applied in all empirical sciences. Typically, **experiments are performed** on an existing system, a human, a desert ant, or a brain region, and the **results are analyzed** in various ways. Often the goal is to develop a **model** to predict the outcome of future experiments.





SYNTHETIC APPROACH

The synthetic approach works by creating **an artificial system** that reproduces certain aspects of a natural system. Rather than focusing on producing the correct experimental results, that is, the correct output, we can try to **reproduce the internal mechanisms** that have led to the particular results → "understanding by building"

From REPRODUCING THE RESULT to WHY THE RESULT CAME ABOUT

Frame of reference problem

The Oxford ethologist David McFarland called it "anthropomorphization: the incurable disease."

Herb Simon's seminal book "The Sciences of the Artificial" (the ant walking along a beach)

Agents can exploit physical laws even if they are not aware of them. Intelligence, in this sense, is not so much a property of an agent or of the brain or of evolution, but rather resides in the eye of the beholder who observes the exploitation.

Traditional A.I.

What is intelligence?

In 1921, the Journal of Educational Psychology asked 14 experts for definitions

Result: 14 different definitions!

Intelligence is difficult to define and understand
Maybe one should try to build artificial intelligent systems in order to understand intelligence

Traditional A.I.

A.I. research is…

"…the science and engineering of making intelligent machines"

(McCarthy, 1955)

"…the art of creating machines that perform functions that require intelligence when performed by people"

(Kurzweil, 1990)

"…the study of how to make computers do things at which, at the moment, people are better"

(Rich & Knight, 1991)

The modern definition of A.I. is *"the study and design of intelligent agents"*

Success of traditional A.I.

What matters in the classical approach is the **algorithm** or the **program** and not the hardware (the body or brain) on which it runs. Abstract functioning that is independent of the specifics of a particular hardware is an extremely powerful idea and constitutes one of the main reasons why computing has conquered the world, so to speak: <u>all that matters are the programs that run on your computer; the hardware is irrelevant</u>.

- **Expert systems** (medical and technical diagnosis, configuration of complex computer systems, commercial loan assessment, and portfolio management)
- **Search engines** on the internet use clever machine learning algorithms
- **Text-processing systems** use algorithms, which try to infer your intentions from the context of what you have done earlier, and will often volunteer advice
- **Data-mining systems** have been developed that heavily rely on machine learning techniques
- **Natural language processing**, interfaces, computer games, and controls for appliances, home electronics, elevators, cars, and trains abound with AI algorithms

human intelligence = information processing ?

Problem of traditional A.I.

However, the original intention of artificial intelligence was not to develop clever algorithms, but to understand intelligence. It is now generally agreed that the classical approach has failed to deepen our understanding of many intelligent processes.
Natural forms of intelligence require a direct interaction with the real world.

How do we make sense of an everyday scene or recognize a face in a crowd?

How do we manipulate objects, especially flexible and soft objects and materials like clothes, string, and paper?

How do we walk, run, ride a bicycle, and dance?

What is common sense all about, and how are we able to understand and produce everyday natural language?

consider not just the brain, but how the body and brain of an intelligent agent interact with the real world

Problem of traditional A.I.

Industrial robotics:
birth and growth of theories and techniques for robot control

Service robotics:
birth and growth of theories and techniques for robot perception & action control

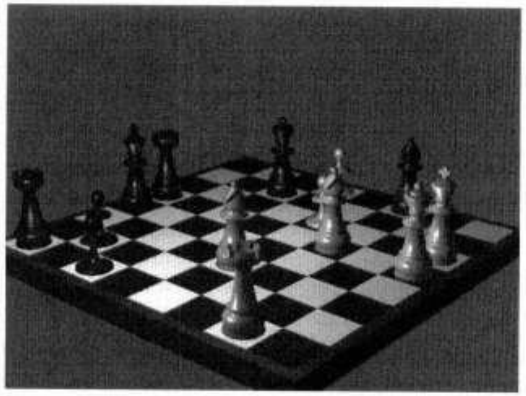
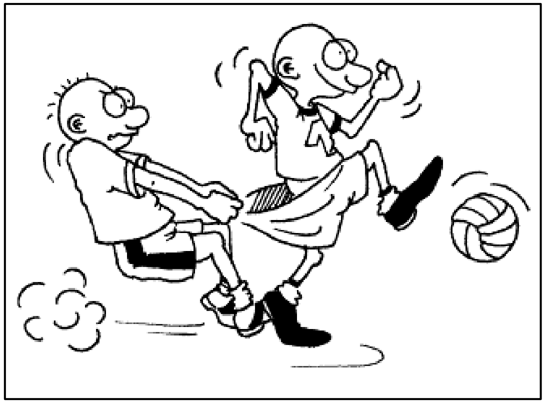
Structured enviroment

Unstructured enviroment

Photo: Center for Robot-Assisted Search and Rescue

Real world vs. virtual world

Classical models focus on high-level processes like problem solving, reasoning, making inferences.

CHESS	SOCCER
	
Formal game	Nonformal game
Virtual world with discrete, clearly defined states, board positions, operations, and legal moves	Real world with no uniquely defined states, continuous and never repeated
Complete information	Information is always incomplete
Static game	Time pressure

The symbol-grounding problem

Symbols are typically defined in a purely syntactic way by how they relate to other symbols and how they are processed by some interpreter.
The relation of symbols (e.g., in database applications) to the outside world is never discussed; it is assumed as somehow given, with the (typically implicit) assumption that designers and potential users know what the symbols mean.

Using symbols in a computer system is no problem as long as there is a human interpreter who can be safely expected to be capable of establishing the appropriate relations to some outside world: **the mapping is "grounded" in the human's experience of his or her interaction with the real world**.

However, once we remove the human interpreter from the loop, as in the case of autonomous agents, we have to take into account that the system needs to interact with the environment on its own.

Design your robot as a complete agent behaving in real world

Note: we use the term agent whenever we do not want to make a distinction between humans, animals, or robots, i.e., when what we say applies to all three.

Complete agents

In laboratory studies, people are often tested on tasks that are not only somewhat artificial but also unusually difficult for humans: subjects are asked, for example, to remember long lists of numbers or to read text upside down.

“if we are to learn something relevant about intelligence—something that holds true in real-world behavior—we need to study complete systems, i.e., systems that have to act and perform tasks autonomously in the real world”

Toda, M. (1982). Man, robot, and society. The Hague: Nijhoff.

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Properties of complete agents

1. *A complete agent is subject to the laws of physics.*

Walking requires energy, friction, and gravity in order to work.

Because the agent is embodied, it is a physical system (biological or not) and thus subject to the laws of physics from which it cannot possibly escape; it must comply with them. If an agent jumps up in the air, gravity will inevitably pull it back to the ground.

Properties of complete agents

2. A complete agent generates sensory stimulation.

When we walk, we generate sensory stimulation, whether we like it or not:

- when we move, objects seem to flow past us (this is known as optic flow);
- by moving we induce wind that we then sense with our skin and our hair;
- walking also produces pressure patterns on our feet;
- and we can feel the regular flexing and relaxing of our muscles as our legs move.

Properties of complete agents

3. *A complete agent affects its environment.*

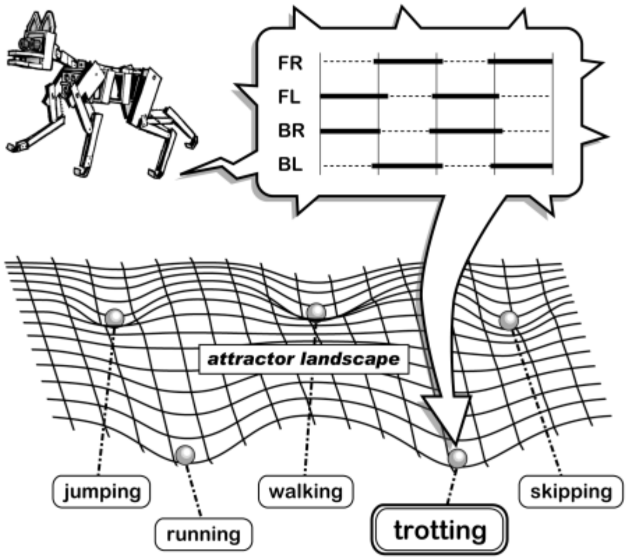
- When we walk across a lawn, the grass is crushed underfoot;
- when we breathe, we blow air into the environment;
- when we walk and burn energy, we heat the environment;
- when we drink from a cup, we reduce the amount of liquid in the glass;
- when we drop a cup, it breaks;
- when we talk, we put pressure waves out into the air;
- when we sit down in a chair, it squeaks, and the cushion is squashed.

Properties of complete agents

4. Agents tend to settle into attractor states.

Agents are dynamical systems, and as such they have a tendency to settle into so called attractor states. Horses, for example, can walk, trot, canter, and gallop, and we or at least experts can clearly identify when the horse is in one of these walking modes, or gaits, the more technical word for these behaviors.

These gaits can be viewed as **attractor states**. The horse is always in one of these states, except for short periods of time when it transitions between two of them, for example from canter to gallop.

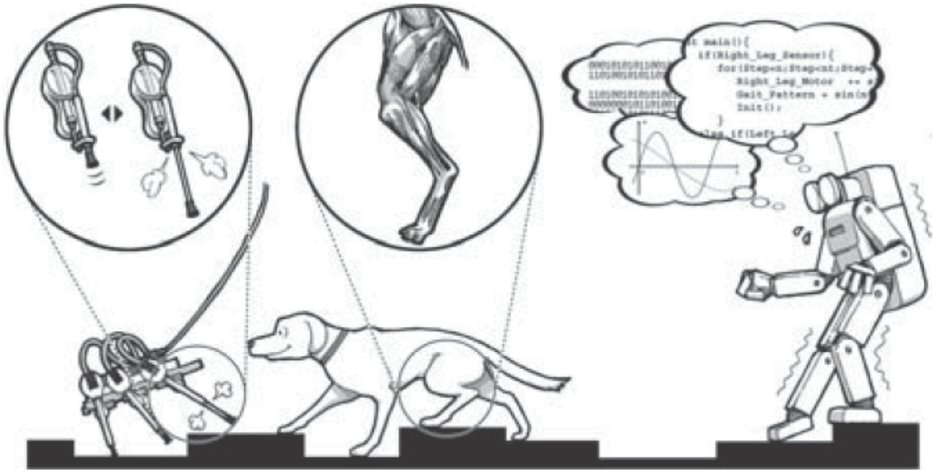


We should point out here that the attractor states into which an agent settles are always the result of the interaction of three systems: the agent's body, its brain (or control system), and its environment.

Properties of complete agents

5. Complete agents perform morphological computation.

By “morphological computation” we mean that certain processes are performed by the body that otherwise would have to be performed by the brain.



An example is the fact that the human leg’s muscles and tendons are elastic so that the knee, when the leg impacts the ground while running, performs small adaptive movements without neural control.

The control is supplied by the muscle tendon system itself, which is part of the morphology of the agent.

Properties of complete agents

1. They are **subject to the laws of physics** (energy dissipation, friction, gravity)

2. They **generate sensory stimulation** through motion and generally through interaction with the real world.

3. They **affect the environment** through behaviour.

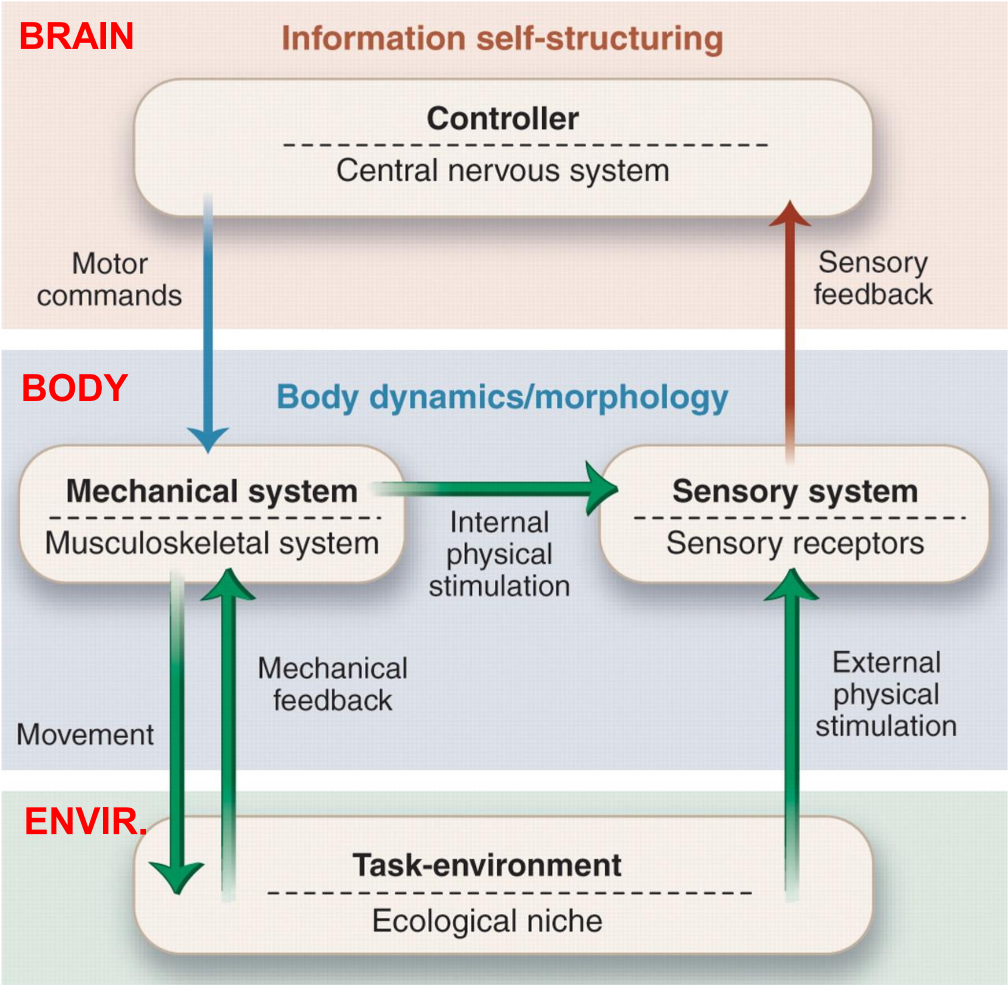
4. They are **complex dynamical systems** which, when they interact with the environment, have attractor states

5. They **perform morphological computation**.

These properties are simply unavoidable consequences of embodiment.

Systems that are not complete hardly ever possess all of these properties. For example, a vision system consisting of a fixed camera and a desktop computer does not generate sensory stimulation because it cannot produce behavior, and it influences the environment only by emitting heat and light from the computer screen. Moreover, it does not perform morphological computation and does not have physical attractor states that could be useful to the system.

Embodiment



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Mechanical feedback describes the reaction that the agent's mechanical system experiences after acting on the environment.

External physical stimulation refers to the effects of the environment on the agent's sensory system.

Internal physical stimuli refer to forces and torques developed in the muscles and joint-supporting ligaments.

The agent is embedded in a **task-environment/ecological niche**. This encompasses all forces such as gravity and friction that act on the agent.

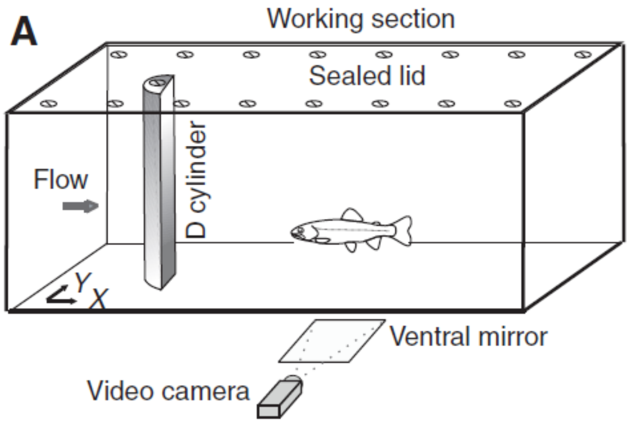
Locomotion case studies

The fact that moving from one place to another, or locomotion, requires a body, comes as no surprise. However, it has been treated predominantly as a control problem by many; the body playing the part of a mere tool that has to be commanded appropriately.

On contrary: shaping the body morphology and thereby the dynamics that result from the interaction with the environment can lead to stable and efficient locomotion, requiring very little control.

Brain-less and power-less creatures

Euthanized trout

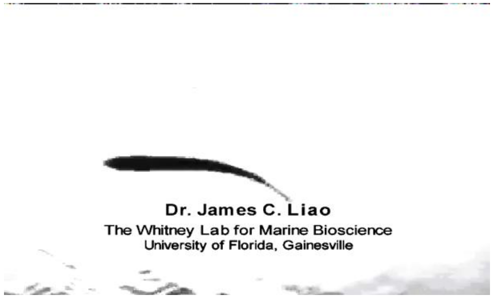


Fish exploit the physical characteristics of their environment by means of their body morphology to reduce energy consumption while swimming

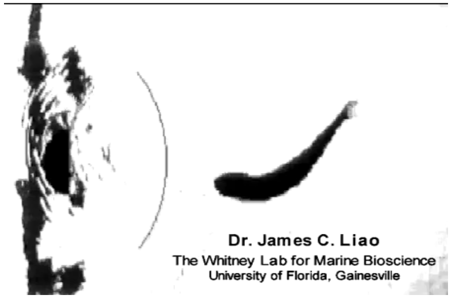
Two hydrodynamic benefits of station holding behind a cylinder:

- simply swim against the current in the region of reduced flow
- generate lift to move against the flow by altering their body kinematics to synchronize with the shed vortices

Live trout in free stream flow



Live trout behind the cylinder



Euthanized trout behind the cylinder

Winner of 2024 Ig Nobel Prize (science that makes people LAUGH, then THINK)

https://youtu.be/ukBwV9Lap2A?si=hktDuTqF5tWz3fsq&t=2483

Brain-less and power-less creatures

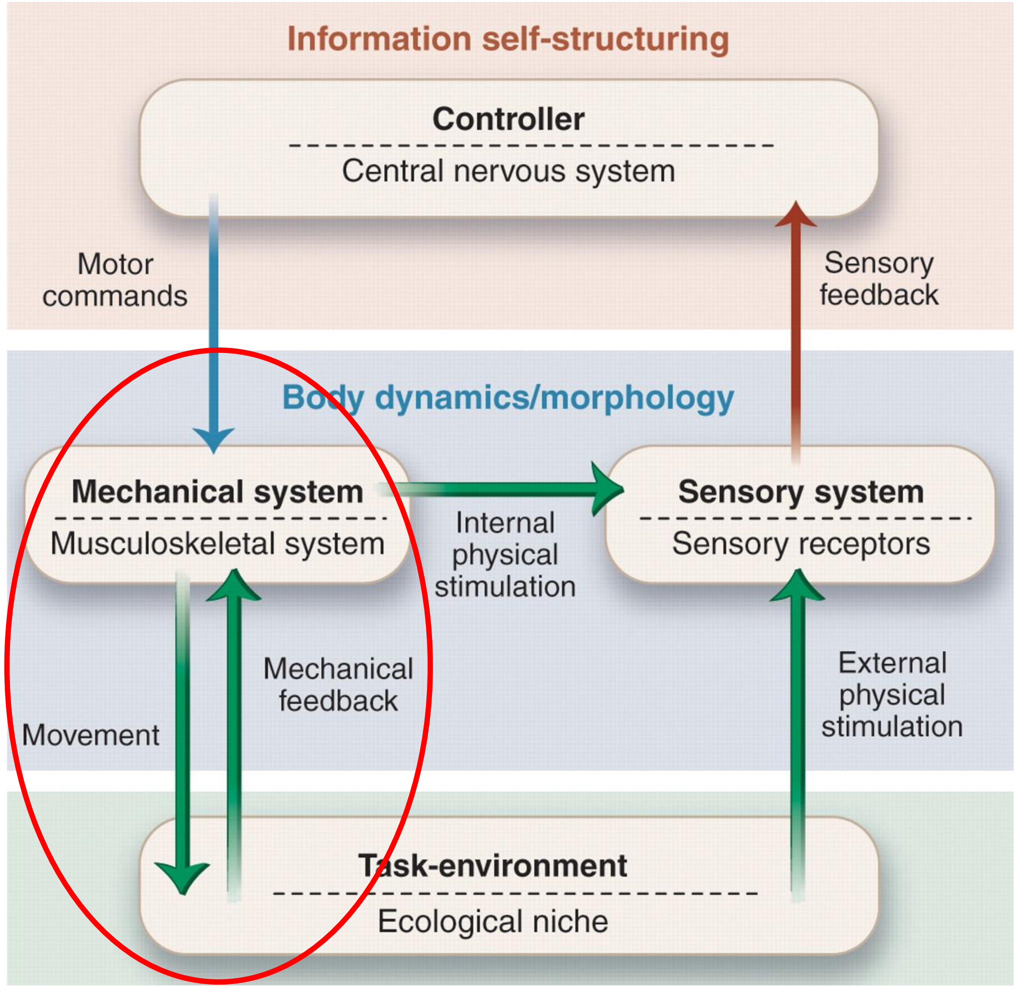
Passive dynamic walkers

The passive dynamic walker, which goes back to McGeer (1990), is capable of walking down an incline without any actuation and without control. In other words, there are no motors, no sensors, and there is no microprocessor on the robot; it is brainless, so to speak. Its locomotion is an outcome of the slope of the incline (gravity is the only power source), and the mechanical parameters of the walker (mainly leg segment lengths, mass distribution, and foot shape).

Locomotion can be realized through pure, but carefully tuned mechanics only

Brain-less and power-less creatures

Theoretical Scheme



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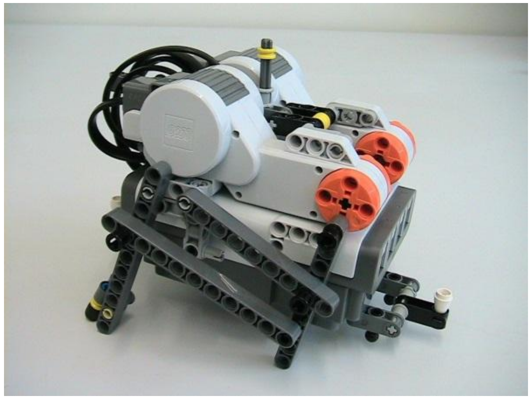
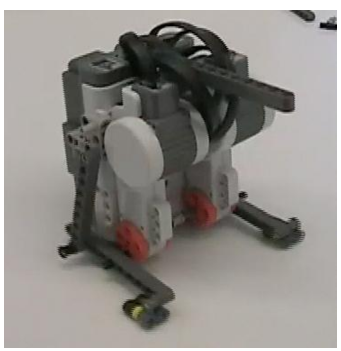
Brain-less and powered creatures

Behavioural diversity - Crazy bird



It only has two motors and no sensors, the two motors are synchronized, turn at the same speed and in the same direction. The front legs are attached eccentrically to the wheels driven by the motors and are physically connected to the hind legs. Four experimental setups were tested with different phase delay between left and right legs (i.e., 0° , 90° , 180° , and 270°). Despite its simplicity, the robot could reach every point on a table by changing a single control parameter (i.e., the phase delay between its legs).

The controller only has to induce a global bias rather than exactly controlling the joint angles or other parameters of the agent’s movement



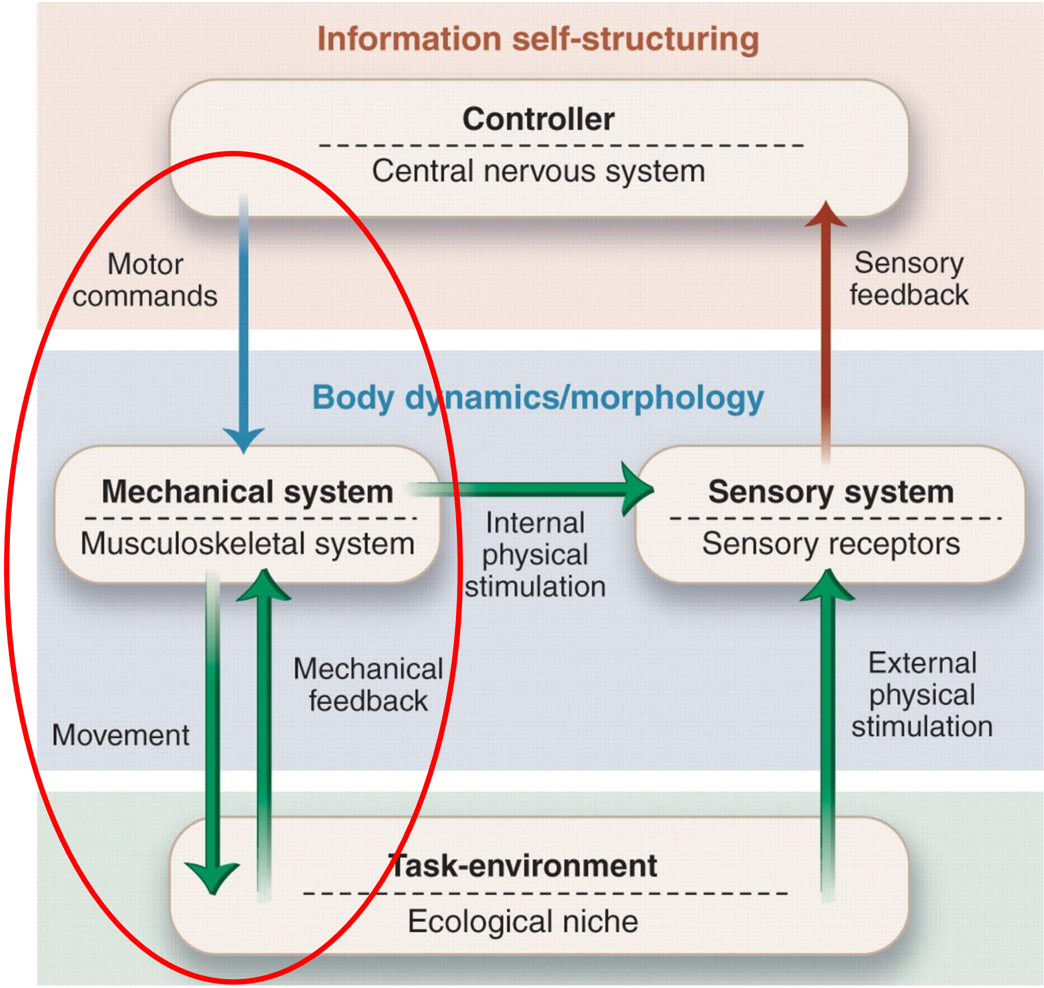
Brain-less and powered creatures

Self-stabilization

Passive dynamic walkers have shown that locomotion can be realized through pure, but carefully tuned mechanics. However, how stable or adaptive is such a solution? In other words, how does a brainless machine cope with different slopes or with disturbances? The theory of nonlinear dynamical systems is often employed to analyze the phenomena involved in the mechanical (and also neural) aspects of locomotion. The walker is an example of a nonlinear dynamical system and walking patterns (which are periodic motions) correspond to limit cycles. Limit cycles in a nonlinear system can display attractive behavior, i.e. nearby trajectories are ‘pulled' toward the limit cycle.

Brain-less and powered creatures

Theoretical Scheme



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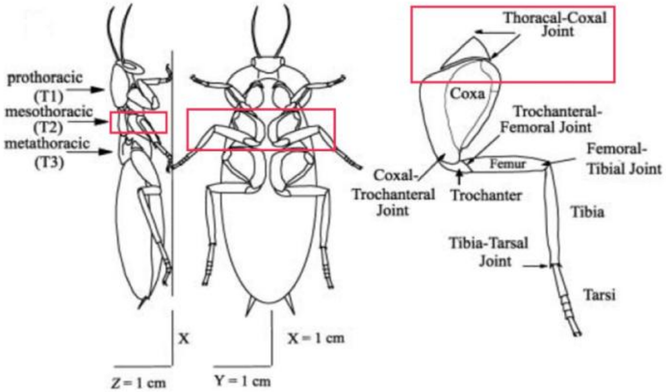
Simple control exploiting body dynamics and interaction with the environment

Cockroaches climbing over obstacles

Cockroaches cannot only walk fast over relatively uneven terrain, but they can also, with great skill, negotiate obstacles that exceed their body height. They have **complex bodies** with wings and three thoracic segments, each one with a pair of legs. Each leg has more than seven degrees of freedom (legs and feet are deformable to various extents), some joints in the legs are very complicated and there is also an intricate set of joints in the feet.



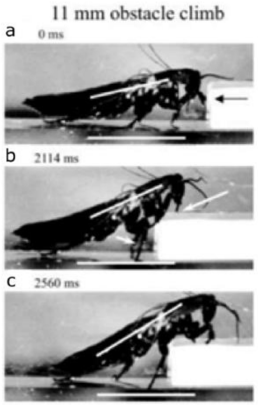
Although both the brain and the thoracic ganglion have a large number of neurons, only around 250 neurons - a very small number - descend from the brain to the body.

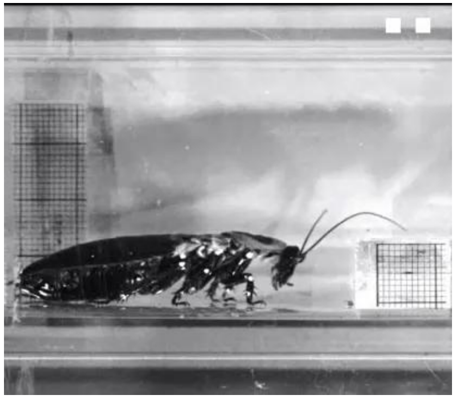
Now, how is it possible to control all these degrees of freedom with such a small number of descending neurons?

Simple control exploiting body dynamics and interaction with the environment

Cockroaches climbing over obstacles

Suppose that the animal is walking on the ground using its local neural circuits which account for stable forward locomotion. If it now encounters an obstacle - whose presence and approximate height it can sense by means of its antennae. The brain, rather than changing the movement pattern, "re-configures" the shoulder joint.





The local neural circuits continue doing the same thing but because of the new configuration of the shoulder joint is different, the effect of the neural circuits on the behavior changes

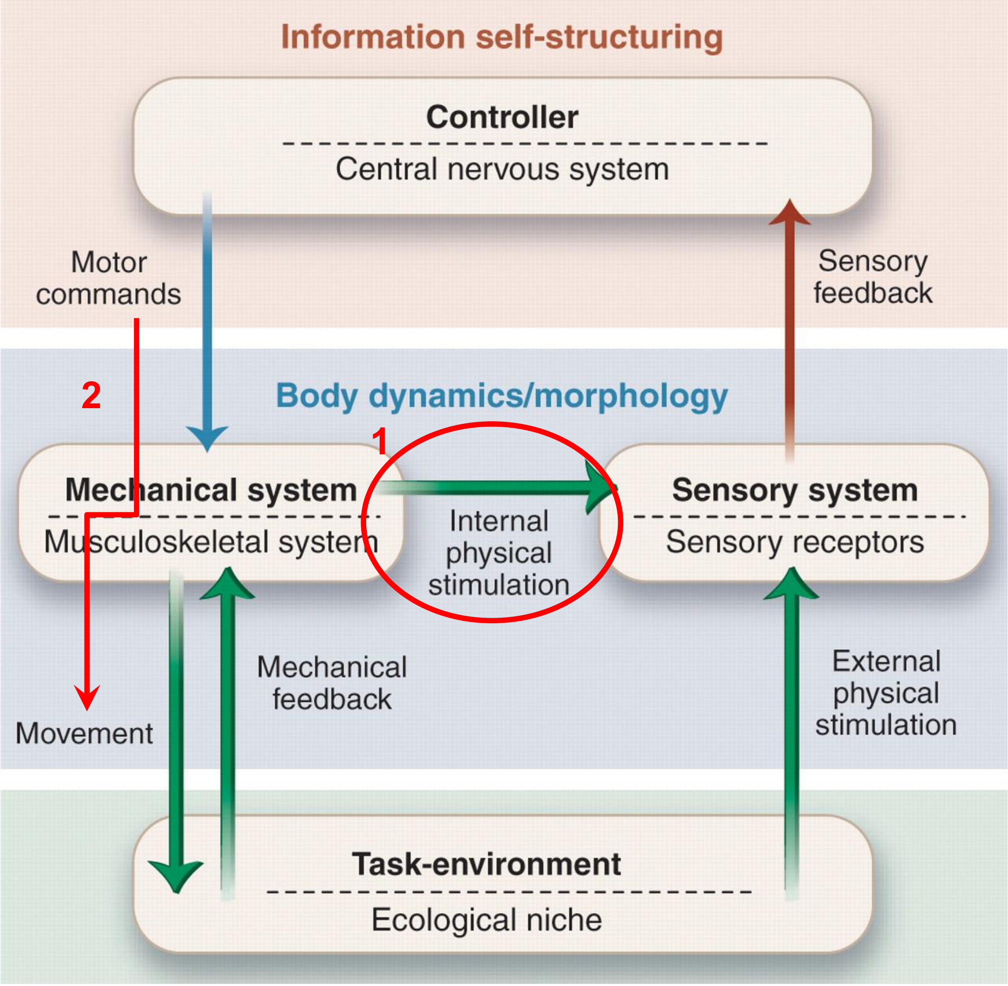
The brain orchestrates the dynamics of locomotion by "outsourcing" some of the functionality to local neural circuits and morphology through an ingenious kind of cooperation with body and decentralized neural controllers.

ADVANTAGES

1. the control problem in climbing over obstacles can be solved **efficiently** with relatively few neurons
2. this is a "cheap solution" because much of what the brain would have to do, is **delegated** to the morphology
3. it takes advantage of the **inherent stability** of the local feedback circuits rather than working in opposition to them
4. it illustrates a new way of designing controllers by **exploiting mechanical change and feedback**

Simple control exploiting body dynamics and interaction with the environment

Theoretical Scheme



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Grasping case studies

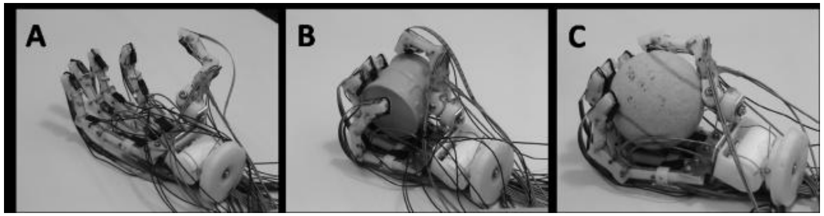
Morphology and materials contribute to grasping behavior. Hand joint structure, muscle mechanics, and the distribution and density of bone to joint movements and muscle recruitment during manipulative behavior are all important variables.

Illustration of ‘cheap grasping', i.e. grasping that is stable and reliable, yet requires little control.

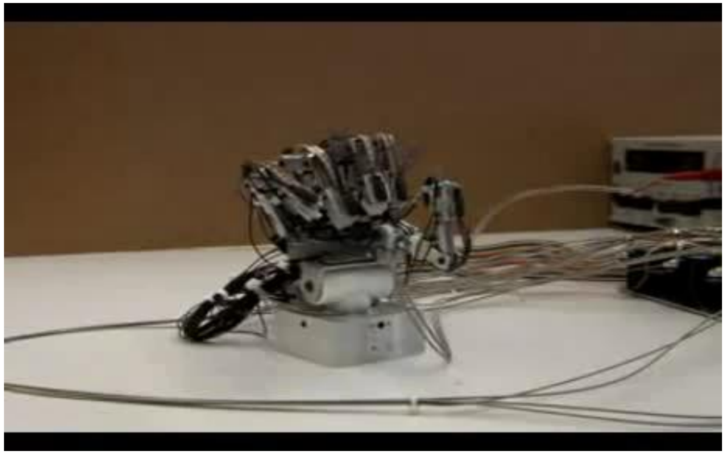
Cheap grasping

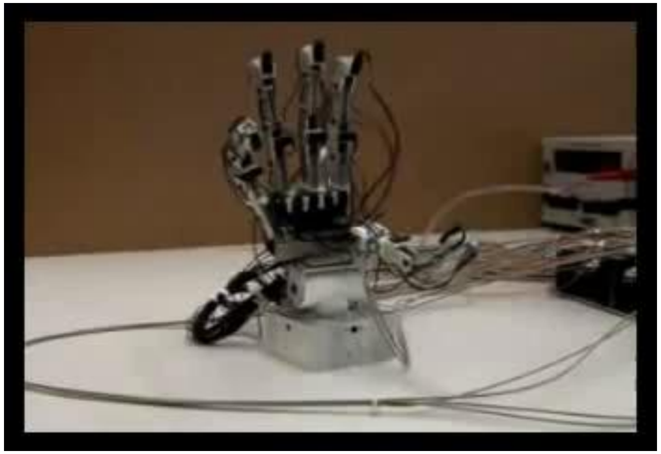
Robotic Yokoi hand

The 18 degrees-of-freedom (DOF) tendon driven hand which can be used as a robotic and a prosthetic hand, is partly built from elastic, flexible, and deformable materials. The tendons are elastic, the finger tips are deformable and between the fingers there is also deformable material. When the hand is closed, the fingers will, because of the anthropomorphic morphology, automatically come together. For grasping an object, a simple control scheme, a ‘close' is applied.



Because of the morphology of the hand, the elastic tendons, and the deformable finger tips, the hand will automatically self-adapt to the object it is grasping



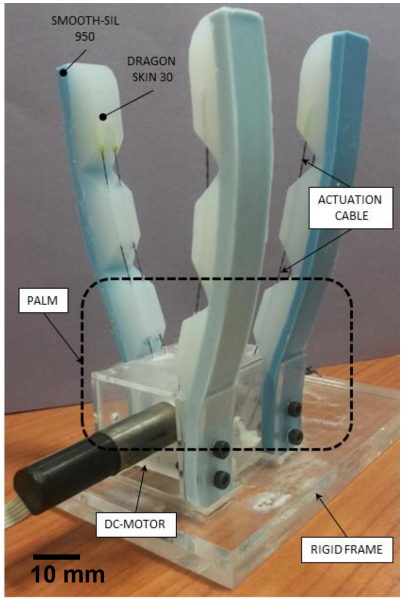


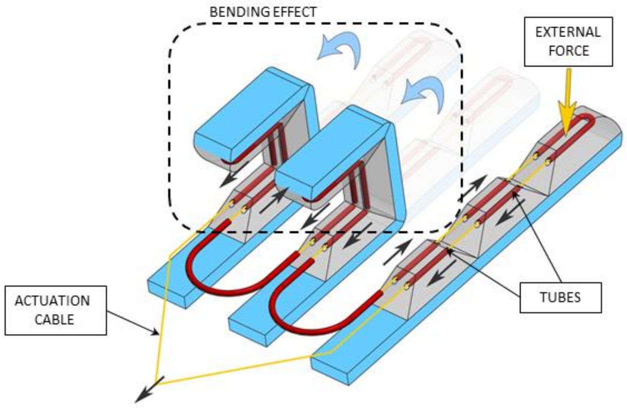
Cheap grasping

SSSA soft robotic gripper

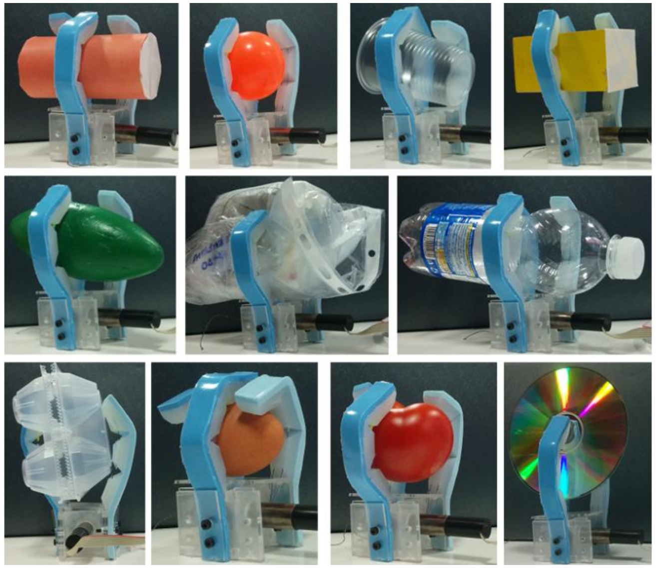
The 9 passive DOF (with 1 active DOF) tendon driven hand which can be used as a robotic gripper, is completely built from elastic, flexible, and deformable materials. The tendons are inelastic, the fingers are also deformable material. When the hand is closed, the fingers will automatically come together because of the anthropomorphic morphology. For grasping an object, a simple control scheme, a 'close' is applied.

Because of the morphology of the hand, and the deformable finger, the hand will automatically self-adapt to the object it is grasping





Underactuated inter- and intra-digital soft mechanism

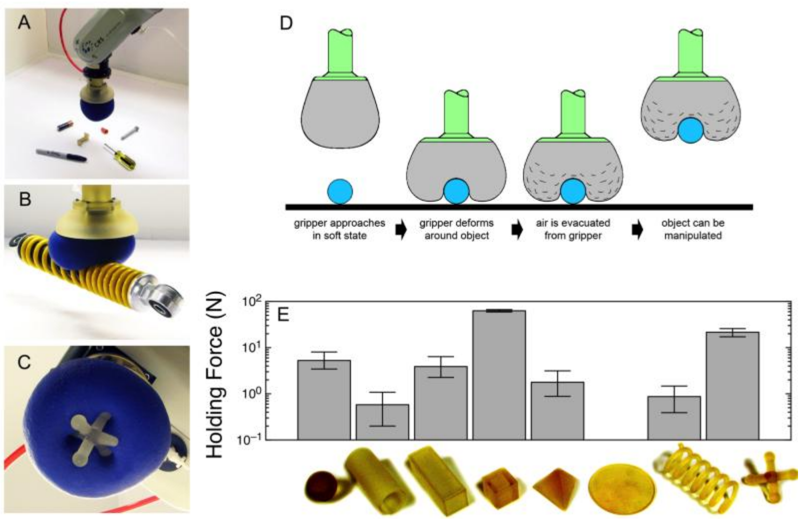


Cheap grasping

Universal gripper

Individual fingers are replaced by a single mass of granular material (e.g., ground coffee). The ‘bag' containing granular material is pressed onto an object, flows around it, and conforms to its shape. Then, a vacuum pump is used to evacuate air from the gripper, which makes the granular material jam and stabilize the grasp.

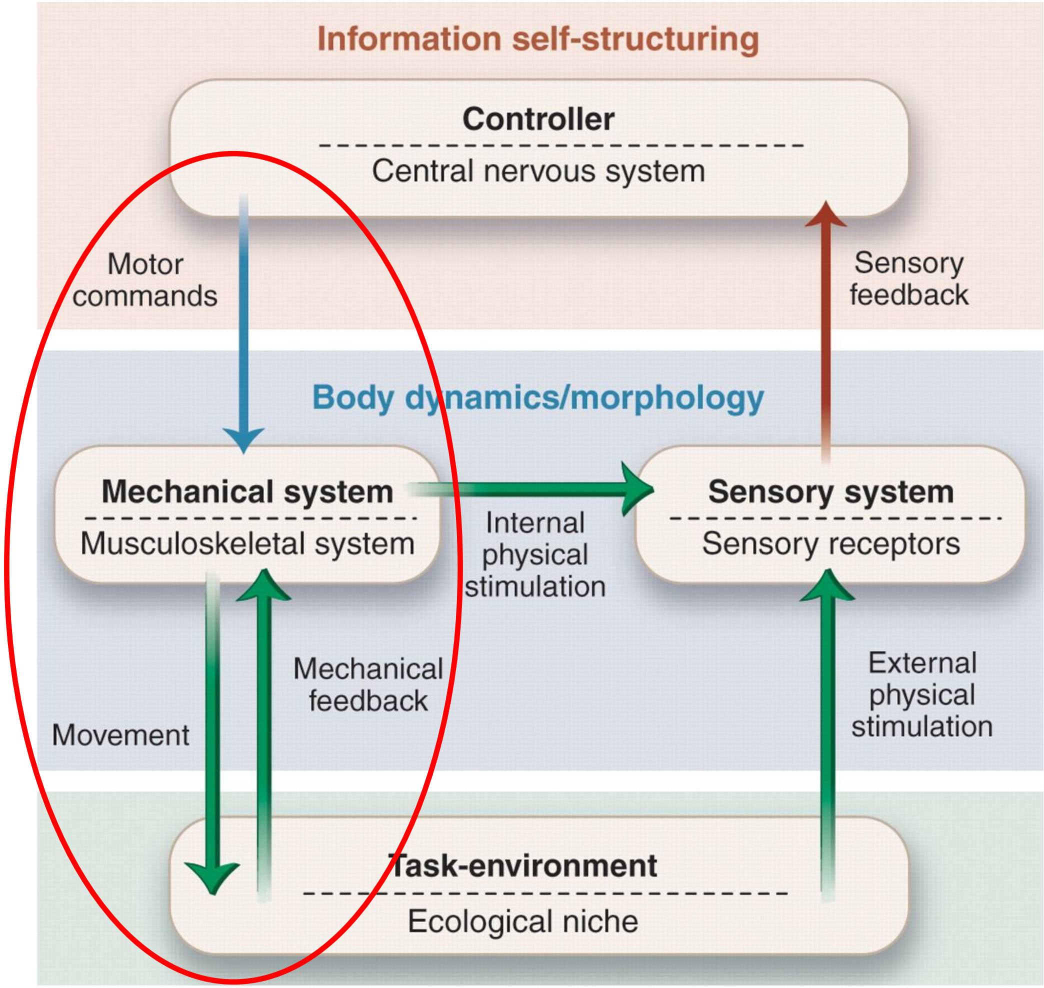
- geometric constraints from interlocking between gripper and object surfaces
- static friction from normal stresses at contact
- an additional suction effect, if the gripper membrane can seal off a portion of the object's surface



The gripper conforms to arbitrary shapes passively, that is without any sensory feedback, thanks to its morphological and mechanical properties only.

Cheap grasping

Theoretical Scheme



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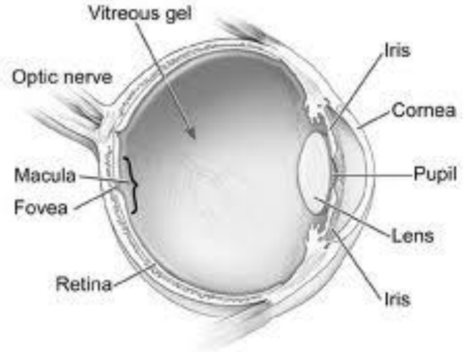
Implications of Embodiment: Information self-structuring

Body and brain should not be viewed as competitors, but rather collaborators. In more complex scenarios, mechanical 'intelligence' has to be aided by software or control. In order for a controller to be able to take the right decisions and issue proper motor commands, it needs to perceive the relevant information regarding the agent's interaction with the environment.

The embodiment shapes and structures the information that reaches the brain (or controller) through:

- morphology of the sensory apparatus
- active generation of information through sensory-motor coordination

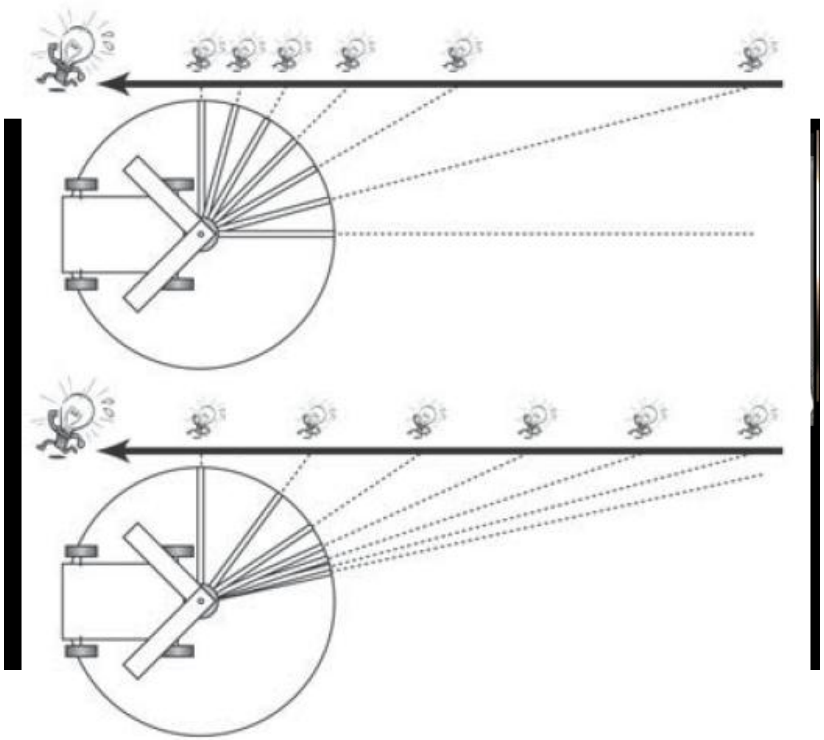
Morphology



Human eye

The retina of a human eye is a variable resolution sensor: the distribution of photoreceptors is non-homogeneous. The density of cones, which are used for high acuity vision, is greatest in the center (fovea). Through this morphological arrangement, a limited number of sensing and processing elements can provide both high acuity in the center of the visual field, and a wide field of view.

Insect eye



For many objectives motion detection is all that is required. Motion detection can often be simplified if the light-sensitive cells are not spaced evenly. In the compound eye of the house fly the spacing of the facets is denser toward the front of the animal. This non-homogeneous arrangement, in a sense, compensates for the phenomenon of motion parallax: it performs the 'morphological computation'. This implies that under the condition of straight flight, the same motion detection circuitry can be employed for motion detection for the entire eye.

Certain aims can be solved by proper morphological arrangement of the sensors without changing anything inside the neural controller

Sensory-motor Coordination

The sensory stimulation is not passively received, but rather actively generated.

"We begin not with a sensory stimulus, but with a sensory-motor coordination [...] In a certain sense it is the movement which is primary, and the sensation which is secondary, the movement of the body, head, and eye muscles determining the quality of what is experienced. In other words, the real beginning is with the act of seeing; it is looking, and not a sensation of light."

(Dewey, 1896)

Visual perception

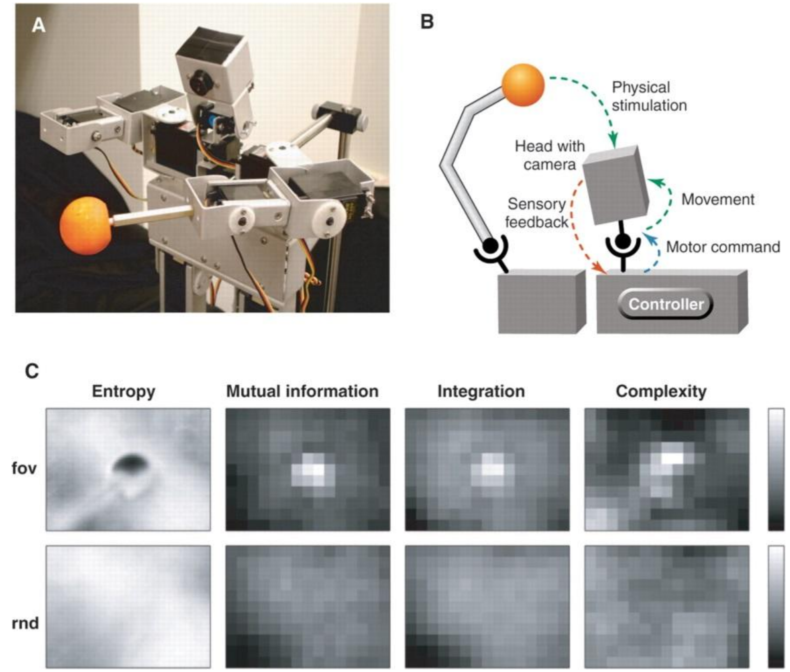
fov: the sensory feedback is exploited by the controller of the robot head with camera to track the end-effector

rnd: the movement of the camera is unrelated to the movement of the ball (sensory-motor coupling is disrupted)

(third with two eyes): one eye tracking the object and the other following its movements, i.e. without vergence

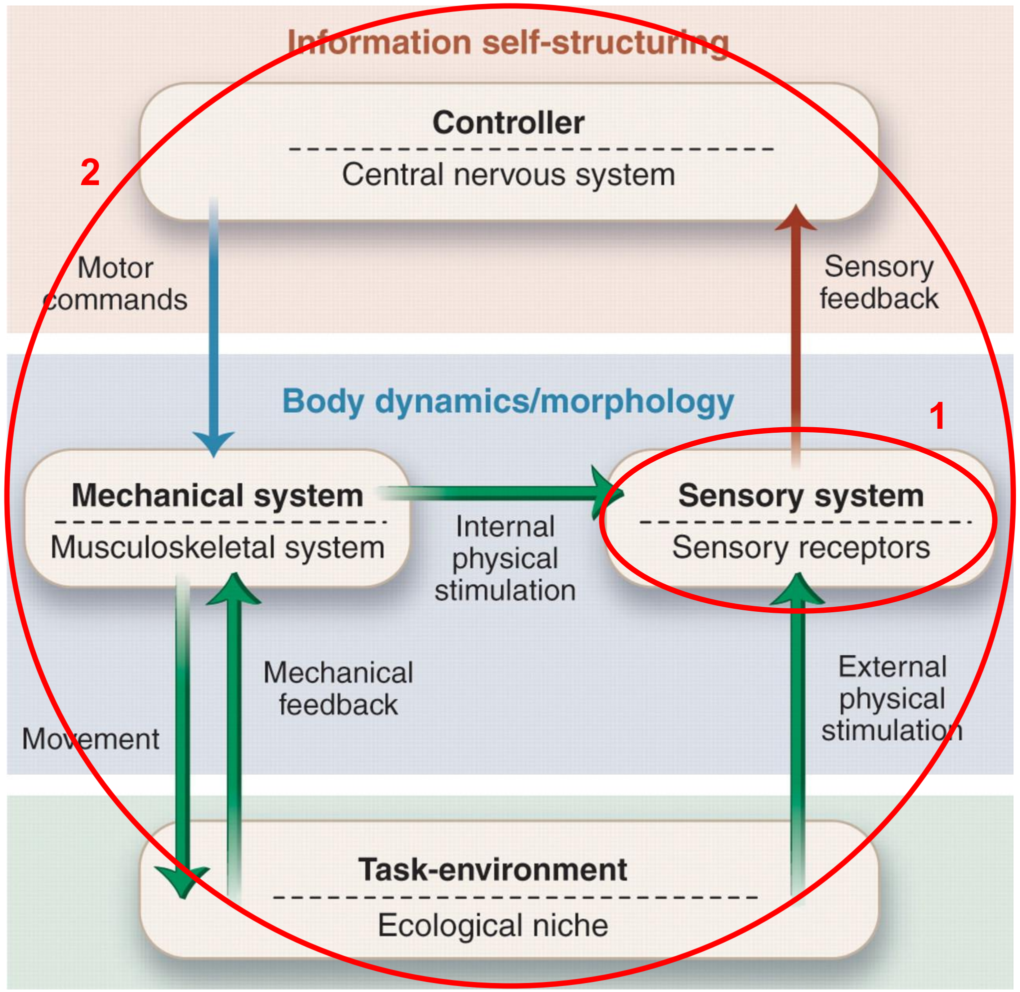
no information structure

Morphology and active perception cannot be considered in isolation.



Information self-structuring

Theoretical Scheme



Motor commands are the instructions generated by the central nervous system / the controller.

Mechanical feedback describes the reaction that the agent's mechanical system experiences after acting on the environment.

External physical stimulation refers to the effects of the environment on the agent's sensory system.

Internal physical stimuli refer to forces and torques developed in the muscles and joint-supporting ligaments.

The agent is embedded in a **task-environment/ecological niche**. This encompasses all forces such as gravity and friction that act on the agent.

How to design a complete agent?

Unlike formal systems, the real world is messy, so to speak, we **cannot expect a clean, axiomatic theory or a set of principles** that logically follow from one another.

So, the set of design principles that we will present is not a formal system, but a **tightly interdependent set of design heuristics** that on the one hand provide guidance on how to go about building agents, and on the other characterize the nature of intelligent systems.

Agent Design Principle 1

The **three-constituents** principle

- define the ecological niche (ENVIRONMENT)
- define the desired behaviour and tasks (TASK)
- design the agent (BODY)

Scaffolding
Given the obvious usefulness of changing the environment to simplify our lives (scaffolding our environment) it is truly surprising that most robots do not significantly change their environments to make their tasks easier!

Agent Design Principle 2

The **complete agent** principle

- think about the complete agent behaving in the real world (everything is tightly coupled)

How to consider the environment in the design workflow?

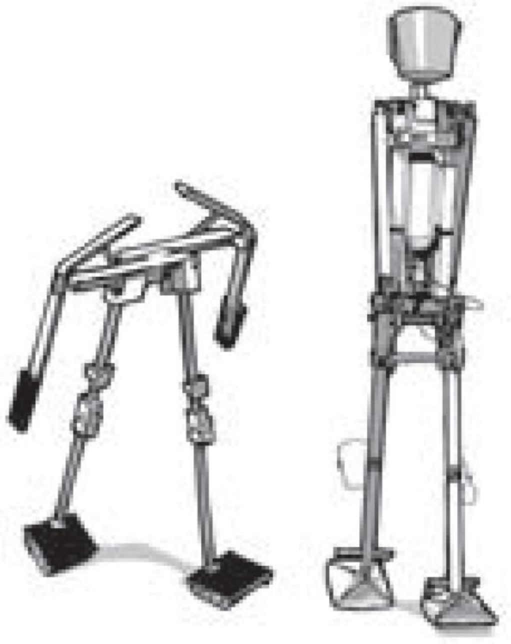
Geometry
Material
Interaction

} Highly nonlinear → FEM simulation vs. reality gap

Agent Design Principle 3

Cheap design

- If agents are built to <u>exploit</u> the properties of the ecological niche and the characteristics of the interaction with the environment, their design and construction will be much easier, or “cheaper”



Passive walker

Agent Design Principle 4

Redundancy

Intelligent agents must be designed in such a way that

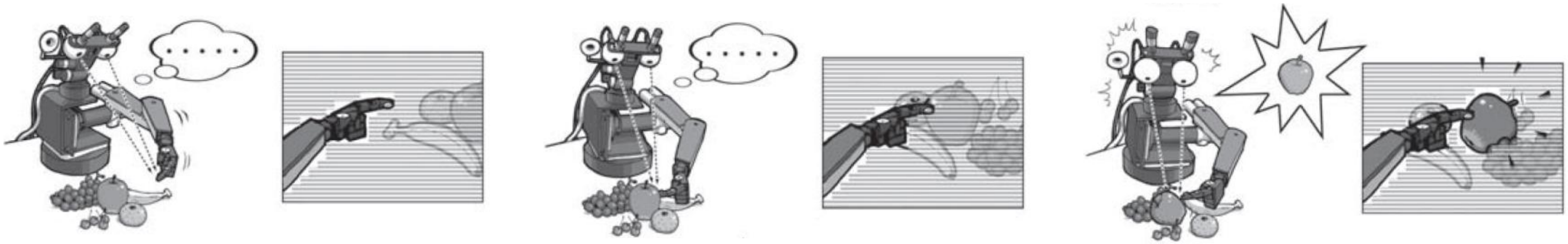
- their different sub systems function on the basis of different physical processes, and
- there is partial overlap of functionality between the different sub systems

Duplication vs. Adaptability and resilience

Agent Design Principle 5

Sensory Motor Coordination

Through sensory motor coordination, structured sensory stimulation is induced as “active perception”



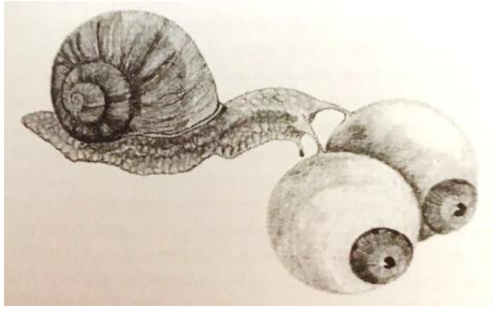
Sensory-motor coordination is always performed with respect to a particular goal or intention.

Agent Design Principle 6

Ecological balance

1. Given a certain task environment, there has to be a match between the <u>complexities</u> of the agent's sensory, motor, and neural systems

2. There is a certain balance or task <u>distribution</u> between morphology, materials, control, and environment.



From *Climbing Mount Improbable* by Dawkins.
A snail with human like, and human sized, eyes. This snail would have a hard time carrying along these giant eyes, but more importantly, they would be only moderately useful, if at all: why bother detecting fast moving predators if you cannot run away from them, or detecting running prey if you are vegetarian? The complexity, weight, and size of the human eyes would only constitute unnecessary baggage, an example of an entirely unbalanced system.

Agent Design Principle 7

Parallel, loosely coupled processes

Intelligence is emergent from a large number of parallel processes that are often coordinated through embodiment, in particular via the embodied interaction with the environment

The sense-think-act model, has proved inappropriate in the real world because

1. it is a one-way model, assuming that sensory stimulation comes first and leads to internal representation
2. with real-time constraints, this way of functioning would simply not be fast enough

Intelligence as a collection of parallel, asynchronous processes that are only loosely coupled. Intelligent behavior is emergent from a large number of such processes.

“Loosely coupled” in contrast to hierarchically coupled processes, but very strongly coupled through embodiment and/or environment interaction (not through the neural/control system).

Agent Design Principle 8

Value

Agents are equipped with a value system which constitutes a basic set of assumptions about what is good for the agent

The value principle is on the one hand very important because it deals with the fundamental issue of what is good for the agent, which then leads to the question of what the agent will or should do in a particular situation. On the other, the value principle is also extremely vague, and there is no consensus in the vast literature about how to approach it, neither in biology and psychology, nor in robotics and artificial intelligence.

References and further reading

Hoffmann, M. & Pfeifer, R. (2011), The implications of embodiment for behavior and cognition: animal and robotic case studies, in W. Tschacher & C. Bergomi, ed., 'The Implications of Embodiment: Cognition and Communication', Exeter: Imprint Academic, pp. 31-58. [arxiv link]

Understanding Intelligence

Rolf Pfeifer and Christian Scheier
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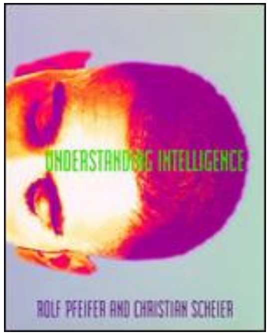

How the Body Shapes the Way We Think
A New View of Intelligence

Rolf Pfeifer and Josh C. Bongard
Foreword by Rodney Brooks, MIT Press 2007